

SESSION 8 • TUTORIAL 2

Tutorial 2: First Law Applied to Closed Systems

Six fully solved problems — processes, cycles, specific heats, paddle work + practice set with answers

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Purpose of this tutorial. To master closed-system energy balances: single processes with mixed signs, complete cycles by the table method, constant-pressure problems using both specific heats, paddle-wheel work, and series-of-process bookkeeping. Six fully solved problems and a practice set are provided.

1. Method Recall

- Write $Q - W = \Delta U$, substitute every interaction with its algebraic sign, and let the algebra fix directions.
- For a cycle, $\Sigma \Delta U = 0$ and therefore $\Sigma Q = \Sigma W$ — the fastest completeness check.
- Constant pressure (ideal gas): $Q = mc_p \Delta T$, $\Delta U = mc_v \Delta T$, $W = Q - \Delta U = mR \Delta T$.
- Shaft/paddle/electrical work enters the balance directly as W (negative when done on the system).

2. Worked Problems

WORKED EXAMPLE T2.1 — Single process, mixed heats

A closed system receives 200 kJ of heat, does 80 kJ of work, and loses 30 kJ of heat to its casing. Find ΔU .

Solution.

1. Net heat $Q = +200 - 30 = +170$ kJ; $W = +80$ kJ.
2. $\Delta U = Q - W = 170 - 80 = +90$ kJ.

ANSWER $\Delta U = +90$ kJ

Note: Sum all heat interactions (with signs) before applying the balance.

WORKED EXAMPLE T2.2 — Cycle of three processes (table method)

A closed system executes a cycle: $Q_{12} = +40$ kJ, $W_{12} = +15$ kJ; $Q_{23} = -10$ kJ, $W_{23} = 0$; $W_{31} = -20$ kJ. Find ΔU for each process and Q_{31} .

Solution.

1. $\Delta U_{12} = 40 - 15 = +25$ kJ; $\Delta U_{23} = -10 - 0 = -10$ kJ.
2. Cycle: $\Sigma \Delta U = 0 \Rightarrow \Delta U_{31} = -(25 - 10) = -15$ kJ.
3. $Q_{31} = \Delta U_{31} + W_{31} = -15 + (-20) = -35$ kJ.
4. Verify: $\Sigma Q = 40 - 10 - 35 = -5$ kJ = $\Sigma W = 15 + 0 - 20 = -5$ kJ ✓ (net work-absorbing cycle).

ANSWER ΔU : +25, -10, -15 kJ | $Q_{31} = -35$ kJ

Process	Q (kJ)	W (kJ)	$\Delta U = Q - W$ (kJ)
1 → 2	+40	+15	+25
2 → 3	-10	0	-10
3 → 1	-35	-20	-15 (so $\Sigma \Delta U = 0$)
Σ cycle	-5	-5	0 ✓

WORKED EXAMPLE T2.3 — Constant-pressure heating of air

2 kg of air ($c_p = 1.005$, $c_v = 0.718$, $R = 0.287$ kJ/kg·K) is heated at constant pressure from 300 K to 450 K. Find Q , ΔU and W .

Solution.

1. $Q = mc_p\Delta T = 2 \times 1.005 \times 150 = 301.5$ kJ.
2. $\Delta U = mc_v\Delta T = 2 \times 0.718 \times 150 = 215.4$ kJ.
3. $W = Q - \Delta U = 86.1$ kJ; check: $mR\Delta T = 2 \times 0.287 \times 150 = 86.1$ kJ ✓.

ANSWER $Q = 301.5$ kJ; $\Delta U = 215.4$ kJ; $W = 86.1$ kJ

Note: The consistency check $W = mR\Delta T$ instantly exposes arithmetic slips.

WORKED EXAMPLE T2.4 — Paddle wheel in an insulated rigid vessel

A rigid, insulated vessel contains 2 kg of air ($c_v = 0.718$ kJ/kg·K). A paddle wheel does 25 kJ of work on the air. Find ΔU and the temperature rise.

Solution.

1. Rigid $\Rightarrow$ no pdV work; insulated $\Rightarrow Q = 0$; paddle work ON the system: $W = -25$ kJ.
2. $\Delta U = 0 - (-25) = +25$ kJ.
3. $\Delta T = \Delta U/(mc_v) = 25/(2 \times 0.718) = 17.4$ K.

ANSWER $\Delta U = +25$ kJ; $\Delta T \approx 17.4$ K

WORKED EXAMPLE T2.5 — Two processes in series

Air passes through states 1 $\rightarrow$ 2 $\rightarrow$ 3. From 1 $\rightarrow$ 2: $Q = +90$ kJ, $W = +35$ kJ. From 2 $\rightarrow$ 3 the process is adiabatic with 40 kJ of work done by the air. Find the overall change in internal energy from 1 to 3.

Solution.

1. $\Delta U_{12} = 90 - 35 = +55$ kJ.
2. $\Delta U_{23} = 0 - 40 = -40$ kJ.
3. $\Delta U_{13} = 55 + (-40) = +15$ kJ (U is a property — changes add).

ANSWER $\Delta U_{13} = +15$ kJ

WORKED EXAMPLE T2.6 — Cooling with simultaneous stirring

While a fluid is stirred with 15 kJ of paddle work, 25 kJ of heat is removed from it. Find ΔU .

Solution.

1. $W = -15$ kJ (on the system); $Q = -25$ kJ.
2. $\Delta U = -25 - (-15) = -10$ kJ.
3. The stirring partly offsets the cooling; the net stored energy still falls.

ANSWER $\Delta U = -10$ kJ

3. Practice Problems (answers given)

#	Problem	Answer
P1	A system receives 120 kJ of heat and 40 kJ of paddle work while doing 65 kJ of expansion work. Find ΔU .	+95 kJ
P2	A cycle has processes A ($Q = +50$ kJ, $W = +20$ kJ) and B ($W = -15$ kJ). Find Q_B and W_{net} .	$Q_B = -45$ kJ; $W_{\text{net}} = +5$ kJ
P3	3 kg of gas ($c_v = 0.72$ kJ/kg·K) cools at constant volume from 200 °C to 80 °C. Find Q .	-259.2 kJ
P4	True or false: in an adiabatic process the internal energy always remains constant. Justify in two lines.	False — $\Delta U = -W$

4. Takeaways

- Substitute with signs; never pre-decide directions.

- Cycle bookkeeping: build the Q-W- ΔU table, close it with $\Sigma \Delta U = 0$, verify $\Sigma Q = \Sigma W$.
- Constant-pressure trio: $Q = mc_p \Delta T$, $\Delta U = mc_v \Delta T$, $W = mR \Delta T$ — and they must satisfy $Q = W + \Delta U$.
- Because U is a property, ΔU over consecutive processes simply adds.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 4 (First Law of Thermodynamics). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.